Balance network loads

ONTAP 9

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Table of Contents

Balance network loads . . . 1
Balance network overview . . . 1
How DNS load balancing works . . . 1
Create a DNS load balancing zone . . . 1
Add or remove a LIF from a load balancing zone . . . 2
Configure DNS services (ONTAP 9.8 and later) . . . 3
Configure DNS services (ONTAP 9.7 and earlier) . . . 6
Configure dynamic DNS services . . . 9

Balance network loads

Balance network overview

You can configure your cluster to serve client requests from appropriately loaded LIFs. This results in a more balanced utilization of LIFs and ports, which in turn allows for better performance of the cluster.

DNS load balancing helps in selecting an appropriately loaded data LIF and balancing user network traffic across all available ports (physical, interface groups, and VLANs).

With DNS load balancing, LIFs are associated with the load balancing zone of an SVM. A site-wide DNS server is configured to forward all DNS requests and return the least-loaded LIF based on the network traffic and the availability of the port resources (CPU usage, throughput, open connections, and so on). DNS load balancing provides the following benefits:

- New client connections balanced across available resources.
- No manual intervention required for deciding which LIFs to use when mounting a particular SVM.
- DNS load balancing supports NFSv3, NFSv4, NFSv4.1, SMB 2.0, SMB 2.1, SMB 3.0, and S3.

How DNS load balancing works

Clients mount an SVM by specifying an IP address (associated with a LIF) or a host name (associated with multiple IP addresses). By default, LIFs are selected by the site-wide DNS server in a round-robin manner, which balances the workload across all LIFs.

Round-robin load balancing can result in overloading some LIFs, so you have the option of using a DNS load balancing zone that handles the host-name resolution in an SVM. Using a DNS load balancing zone, ensures better balance of the new client connections across available resources, leading to improved performance of the cluster.

A DNS load balancing zone is a DNS server inside the cluster that dynamically evaluates the load on all LIFs and returns an appropriately loaded LIF. In a load balancing zone, DNS assigns a weight (metric), based on the load, to each LIF.

Every LIF is assigned a weight based on its port load and CPU utilization of its home node. LIFs that are on less-loaded ports have a higher probability of being returned in a DNS query. Weights can also be manually assigned.

Create a DNS load balancing zone

You can create a DNS load balancing zone to facilitate the dynamic selection of a LIF based on the load, that is, the number of clients mounted on a LIF. You can create a load balancing zone while creating a data LIF.

Before you begin

The DNS forwarder on the site-wide DNS server must be configured to forward all requests for the load balancing zone to the configured LIFs.

The Knowledgebase article How to set up DNS load balancing in Cluster-Mode on the NetApp Support Site contains more information about configuring DNS load balancing using conditional forwarding.

About this task

- Any data LIF can respond to DNS queries for a DNS load balancing zone name.
- A DNS load balancing zone must have a unique name in the cluster, and the zone name must meet the following requirements:
  - It should not exceed 256 characters.
  - It should include at least one period.
  - The first and the last character should not be a period or any other special character.
  - It cannot include any spaces between characters.
  - Each label in the DNS name should not exceed 63 characters.

    A label is the text appearing before or after the period. For example, the DNS zone named storage.company.com has three labels.

Step

Use the `network interface create` command with the `dns-zone` option to create a DNS load balancing zone.

If the load balancing zone already exists, the LIF is added to it. For more information about the command, see ONTAP 9 commands.

The following example demonstrates how to create a DNS load balancing zone named storage.company.com while creating the LIF `lif1`:

```
network interface create -vserver vs0 -lif lif1 -home-node node1
-home-port e0c -address 192.0.2.129 -netmask 255.255.255.128 -dns-zone
storage.company.com
```

Add or remove a LIF from a load balancing zone

You can add or remove a LIF from the DNS load balancing zone of a virtual machine (SVM). You can also remove all the LIFs simultaneously from a load balancing zone.

Before you begin

- All the LIFs in a load balancing zone should belong to the same SVM.
- A LIF can be a part of only one DNS load balancing zone.
- Failover groups for each subnet must have been set up, if the LIFs belong to different subnets.

About this task

A LIF that is in the administrative down status is temporarily removed from the DNS load balancing zone. When the LIF returns to the administrative up status, the LIF is automatically added to the DNS load balancing zone.

Step

Add a LIF to or remove a LIF from a load balancing zone:

If you want to…	Enter…
Add a LIF	`network interface modify -vserver` *`vserver_name`* `-lif` *`lif_name`* `-dns-zone` *`zone_name`* Example: `network interface modify -vserver vs1 -lif data1 -dns -zone cifs.company.com`

Remove a single LIF	`network interface modify -vserver` *`vserver_name`* `-lif` *`lif_name`* `-dns-zone none`
	Example:
	`network interface modify -vserver vs1 -lif data1 -dns -zone none`

Remove all LIFs	`network interface modify -vserver` *`vserver_name`* `-lif * -dns-zone none`
	Example:
	`network interface modify -vserver vs0 -lif * -dns-zone none`
	You can remove an SVM from a load balancing zone by removing all the LIFs in the SVM from that zone.

Configure DNS services (ONTAP 9.8 and later)

You must configure DNS services for the SVM before creating an NFS or SMB server. Generally, the DNS name servers are the Active Directory-integrated DNS servers for the domain that the NFS or SMB server will join.

About this task

Active Directory-integrated DNS servers contain the service location records (SRV) for the domain LDAP and domain controller servers. If the SVM cannot find the Active Directory LDAP servers and domain controllers, NFS or SMB server setup fails.

SVMs use the hosts name services ns-switch database to determine which name services to use and in which order when looking up information about hosts. The two supported name services for the hosts database are files and dns.

You must ensure that dns is one of the sources before you create the SMB server.

> To view the statistics for DNS name services for the mgwd process and SecD process, use the Statistics UI.

Steps

1. Determine what the current configuration is for the hosts name services database. In this example, the hosts name service database uses the default settings.

   `vserver services name-service` *`ns-switch`* `show -vserver` *`vs1`* `-database` *`hosts`*

```
Vserver: vs1
Name Service Switch Database: hosts
Vserver: vs1 Name Service Switch Database: hosts
Name Service Source Order: files, dns
```

2. Perform the following actions, if required.
   a. Add the DNS name service to the hosts name service database in the desired order, or reorder the sources.

      In this example, the hosts database is configured to use DNS and local files in that order.

      `vserver services name-service ns-switch modify -vserver vs1 -database hosts -sources dns,files`

   b. Verify that the name services configuration is correct.

      `vserver services name-service ns-switch show -vserver vs1 -database hosts`

      ```
      Vserver: vs1
      Name Service Switch Database: hosts
      Name Service Source Order: dns, files
      ```

3. Configure DNS services.

   `vserver services name-service dns create -vserver vs1 -domains example.com,example2.com -name-servers 10.0.0.50,10.0.0.51`

   > The vserver services name-service dns create command performs an automatic configuration validation and reports an error message if ONTAP is unable to contact the name server.

4. Verify that the DNS configuration is correct and that the service is enabled.

   ```
   Vserver: vs1
   Domains: example.com, example2.com Name Servers: 10.0.0.50, 10.0.0.51
   Enable/Disable DNS: enabled Timeout (secs): 2
   Maximum Attempts: 1
   ```

5. Validate the status of the name servers.

   `vserver services name-service dns check -vserver vs1`

```
Vserver  Name Server  Status  Status Details
vs1      10.0.0.50    up      Response time (msec): 2
vs1      10.0.0.51    up      Response time (msec): 2
```

Configure dynamic DNS on the SVM

If you want the Active Directory-integrated DNS server to dynamically register the DNS records of an NFS or SMB server in DNS, you must configure dynamic DNS (DDNS) on the SVM.

Before you begin

DNS name services must be configured on the SVM. If you are using secure DDNS, you must use Active Directory-integrated DNS name servers and you must have created either an NFS or SMB server or an Active Directory account for the SVM.

About this task

The specified fully qualified domain name (FQDN) must be unique:

The specified fully qualified domain name (FQDN) must be unique:

- For NFS, the value specified in `-vserver-fqdn` as part of the `vserver services name-service dns dynamic-update` command becomes the registered FQDN for the LIFs.
- For SMB, the values specified as the CIFS server NetBIOS name and the CIFS server fully qualified domain name become the registered FQDN for the LIFs. This is not configurable in ONTAP. In the following scenario, the LIF FQDN is "CIFS_VS1.EXAMPLE.COM":

```
cluster1::> cifs server show -vserver vs1

                                          Vserver: vs1
                        CIFS Server NetBIOS Name: CIFS_VS1
                    NetBIOS Domain/Workgroup Name: EXAMPLE
                      Fully Qualified Domain Name: EXAMPLE.COM
                              Organizational Unit: CN=Computers
Default Site Used by LIFs Without Site Membership:
                                   Workgroup Name: -
                                   Kerberos Realm: -
                             Authentication Style: domain
                CIFS Server Administrative Status: up
                          CIFS Server Description:
                          List of NetBIOS Aliases: -
```

To avoid a configuration failure of an SVM FQDN that is not compliant to RFC rules for DDNS updates, use an FQDN name that is RFC compliant. For more information, see RFC 1123.

Steps

1. Configure DDNS on the SVM:

```
vserver services name-service dns dynamic-update modify -vserver vserver_name
-is- enabled true [-use-secure {true|false} -vserver-fqdn
FQDN_used_for_DNS_updates

vserver services name-service dns dynamic-update modify -vserver vs1 -is
-enabled true - use-secure true -vserver-fqdn vs1.example.com
```

Asterisks cannot be used as part of the customized FQDN. For example, `*.netapp.com` is not valid.

2. Verify that the DDNS configuration is correct:

```
vserver services name-service dns dynamic-update show
```

```
Vserver  Is-Enabled Use-Secure Vserver FQDN      TTL
-------- ---------- ---------- ----------------- -------
vs1      true       true       vs1.example.com   24h
```

Configure DNS services (ONTAP 9.7 and earlier)

You must configure DNS services for the SVM before creating an NFS or SMB server. Generally, the DNS name servers are the Active Directory-integrated DNS servers for the domain that the NFS or SMB server will join.

About this task

Active Directory-integrated DNS servers contain the service location records (SRV) for the domain LDAP and domain controller servers. If the SVM cannot find the Active Directory LDAP servers and domain controllers, NFS or SMB server setup fails.

SVMs use the hosts name services ns-switch database to determine which name services to use and in which order when looking up information about hosts. The two supported name services for the hosts database are `files` and `dns`.

You must ensure that `dns` is one of the sources before you create the SMB server.

> To view the statistics for DNS name services for the mgwd process and SecD process, use the Statistics UI.

Steps

1. Determine what the current configuration is for the `hosts` name services database.

   In this example, the hosts name service database uses the default settings.

```
vserver services name-service ns-switch show -vserver vs1 -database hosts
```

```
                      Vserver: vs1
Name Service Switch Database: hosts
   Name Service Source Order: files, dns
```

2. Perform the following actions, if required.
   a. Add the DNS name service to the hosts name service database in the desired order, or reorder the sources.

      In this example, the hosts database is configured to use DNS and local files in that order.

      ```
      vserver services name-service ns-switch modify -vserver vs1 -database hosts
      -sources dns,files
      ```

   b. Verify that the name services configuration is correct.

      ```
      vserver services name-service ns-switch show -vserver vs1 -database hosts
      ```

3. Configure DNS services.

   ```
   vserver services name-service dns create -vserver vs1 -domains
   example.com,example2.com -name-servers 10.0.0.50,10.0.0.51
   ```

   > The vserver services `name-service dns create` command performs an automatic configuration validation and reports an error message if ONTAP is unable to contact the name server.

4. Verify that the DNS configuration is correct and that the service is enabled.

   ```
              Vserver: vs1
              Domains: example.com, example2.com Name
              Servers: 10.0.0.50, 10.0.0.51
   Enable/Disable DNS: enabled Timeout (secs): 2
     Maximum Attempts: 1
   ```

5. Validate the status of the name servers.

   ```
   vserver services name-service dns check -vserver vs1
   ```

   ```
   Vserver  Name Server  Status  Status Details
   -------  -----------  ------  -----------------------
   vs1      10.0.0.50    up      Response time (msec): 2
   vs1      10.0.0.51    up      Response time (msec): 2
   ```

Configure dynamic DNS on the SVM

If you want the Active Directory-integrated DNS server to dynamically register the DNS records of an NFS or SMB server in DNS, you must configure dynamic DNS (DDNS) on the SVM.

Before you begin

DNS name services must be configured on the SVM. If you are using secure DDNS, you must use Active Directory-integrated DNS name servers and you must have created either an NFS or SMB server or an Active

Directory account for the SVM.

About this task

The specified fully qualified domain name (FQDN) must be unique:

- For NFS, the value specified in `-vserver-fqdn` as part of the `vserver services name-service dns dynamic-update` command becomes the registered FQDN for the LIFs.
- For SMB, the values specified as the CIFS server NetBIOS name and the CIFS server fully qualified domain name become the registered FQDN for the LIFs. This is not configurable in ONTAP. In the following scenario, the LIF FQDN is "CIFS_VS1.EXAMPLE.COM":

```
cluster1::> cifs server show -vserver vs1

                                            Vserver: vs1
                        CIFS Server NetBIOS Name: CIFS_VS1
                    NetBIOS Domain/Workgroup Name: EXAMPLE
                      Fully Qualified Domain Name: EXAMPLE.COM
                              Organizational Unit: CN=Computers
Default Site Used by LIFs Without Site Membership:
                                   Workgroup Name: -
                                   Kerberos Realm: -
                              Authentication Style: domain
                CIFS Server Administrative Status: up
                          CIFS Server Description:
                          List of NetBIOS Aliases: -
```

To avoid a configuration failure of an SVM FQDN that is not compliant to RFC rules for DDNS updates, use an FQDN name that is RFC compliant. For more information, see RFC 1123.

Steps

1. Configure DDNS on the SVM:

   `vserver services name-service dns dynamic-update modify -vserver` *`vserver_name`* `-is- enabled` *`true`* `[-use-secure {true|false} -vserver-fqdn` *`FQDN_used_for_DNS_updates`*

   `vserver services name-service dns dynamic-update modify -vserver` *`vs1`* `-is -enabled` *`true`* `- use-secure` *`true`* `-vserver-fqdn` *`vs1.example.com`*

   Asterisks cannot be used as part of the customized FQDN. For example, `*.netapp.com` is not valid.

2. Verify that the DDNS configuration is correct:

   `vserver services name-service dns dynamic-update show`

```
Vserver  Is-Enabled Use-Secure Vserver FQDN      TTL
-------- ---------- ---------- ----------------- -------
vs1      true       true       vs1.example.com   24h
```

Configure dynamic DNS services

If you want the Active Directory-integrated DNS server to dynamically register the DNS records of an NFS or SMB server in DNS, you must configure dynamic DNS (DDNS) on the SVM.

Before you begin

DNS name services must be configured on the SVM. If you are using secure DDNS, you must use Active Directory-integrated DNS name servers and you must have created either an NFS or SMB server or an Active Directory account for the SVM.

About this task

The specified FQDN must be unique.

> To avoid a configuration failure of an SVM FQDN that is not compliant to RFC rules for DDNS updates, use an FQDN name that is RFC compliant.

Steps

1. Configure DDNS on the SVM:

   `vserver services name-service dns dynamic-update modify -vserver` *`vserver_name`* `-is- enabled` *`true`* `[-use-secure {true|false} -vserver-fqdn` *`FQDN_used_for_DNS_updates`*

   `vserver services name-service dns dynamic-update modify -vserver` *`vs1`* `-is -enabled` *`true`* `- use-secure` *`true`* `-vserver-fqdn` *`vs1.example.com`*

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Vserver  Is-Enabled Use-Secure Vserver FQDN      TTL
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vs1      true       true       vs1.example.com   24h
```

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